



Seep Pahuja
Biology Educator
(NEET UG)



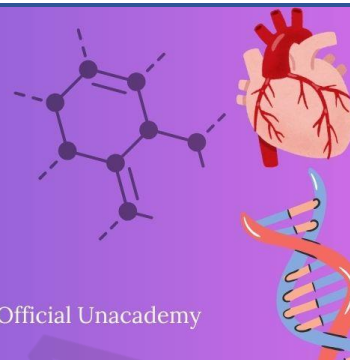
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LAST 10 YEARS

NEET PYQ



Chapter 7 : Structural Organisation in Animals

1. Which of the following statements is correct about the type of junction and their role in our body?

- (1) Adhering junctions facilitate the cells to communicate with each other. **[Re-NEET-2024]**
- (2) Tight junctions help to stop substances from leaking across a tissue.
- (3) Tight junctions help to perform cementing to keep neighbouring cells together.
- (4) Gap junctions help to create gap between the cells and tissues.

2. Which of the following is/are present in female cockroach? **[Re-NEET-2024]**

- A. Collateral gland
- B. Mushroom gland
- C. Spermatheca
- D. Anal style
- E. Phallic gland

Choose the most appropriate answer from the options given below:

- (1) B and D only
- (2) B and E only
- (3) A only
- (4) A and C only

3. In which of the following connective tissues, the cells secrete fibres of collagen or elastin?

- A. Cartilage **[Re-NEET-2024]**
- B. Bone
- C. Adipose tissue
- D. Blood
- E. Areolar tissue

Choose the most appropriate answer from the options given below :

- (1) D, C, D and E only
- (2) A, D, C and E only
- (3) B, C and D only
- (4) A, C and D only

4. Match List I with List II :

[Re-NEET-2024]

List - I

List - II

- A. Squamous Epithelium
- B. Ciliated Epithelium
- C. Glandular Epithelium
- D. Compound Epithelium

- I. Goblet cells of alimentary canal
- II. Inner lining of pancreatic ducts
- III. Walls of blood vessels
- IV. Inner surface of Fallopian tubes

Choose the correct answer from the options given below :

(1) A-II, B-III, C-I, D-IV

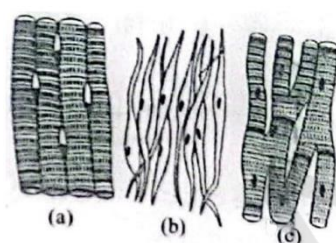
(2) A-II, B-IV, C-III, D-I

(3) A-III, B-I, C-II, D-IV

(4) A-III, B-IV, C-I, D-II

5. Three types of muscles are given as a, b and c. Identify the correct matching pair along with their location in human body:

[NEET-2024]



Name of muscle/location

(1) (a) Smooth - Toes (b) Skeletal - Legs (c) Cardiac - Heart

(2) (a) Skeletal - Triceps (b) Smooth - Stomach (c) Cardiac - Heart

(3) (a) Skeletal - Biceps (b) Involuntary - Intestine (c) Smooth - Heart

(4) (a) Involuntary - Nose tip (b) Skeletal - Bone (c) Cardiac - Heart

6. In both sexes of cockroach, a pair of jointed filamentous structures called anal cerci are present on

(1) 5th segment

(2) 10th segment

[NEET-2024]

(3) 8th and 9th segment

(4) 11th segment

7. Match List I with List II:

[NEET-2024]

List I

List II

- A. Unicellular glandular epithelium
- B. Compound epithelium
- C. Multicellular glandular epithelium
- D. Endocrine glandular epithelium

- I. Salivary glands
- II. Pancreas
- III. Goblet cells of alimentary canal
- IV. Moist surface of buccal cavity

(1) A-II, B-I, C-III, D-IV

(2) A-IV, B-III, C-I, D-II

(3) A-III, B-IV, C-I, D-II

(4) A-II, B-I, C-IV, D-III

8. Match List I with List II related to digestive system of cockroach.

[NEET-2024]

List I

- A. The structures used for storing of food
- B. Ring of 6-8 blind tubules at junction of foregut and midgut.
- C. Ring of 100-150 yellow coloured thin filaments at junction of midgut and hindgut.
- D. The structures used for grinding the food.

List II

- I. Gizzard
- II. Gastric Caeca
- III. Malpighian tubules
- IV. Crop

Choose the correct answer from the options given below:

- (1) A-IV, B-II, C-III, D-I
- (2) A-I, B-II, C-III, D-IV
- (3) A-IV, B-III, C-II, D-I
- (4) A-III, B-II, C-IV, D-I

9. Given below are two statements:

[NEET-2023-MANIPUR]

Statement I : Goblet cells are unicellular glands.

Statement II : Earwax is the secretion of exocrine gland.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both Statement I and Statement II are true
- (2) Both Statement I and Statement II are false
- (3) Statement I is true but Statement II is false
- (4) Statement I is false but Statement II is true

10. The Cockroach is :

[NEET-2023-MANIPUR]

- (1) Ureotelic only
- (2) Ureotelic and Uricotelic
- (3) Ammonotelic only
- (4) Uricotelic only

11. Given below are two statements :

[NEET-2023-MANIPUR]

Statement I : In cockroach, the forewings are transparent and prothoracic in origin.

Statement II : In cockroach, the hind wings are opaque, leathery and mesothoracic in origin.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both Statement I and Statement II are true
- (2) Both Statement I and Statement II are false
- (3) Statement I is correct but Statement II is false
- (4) Statement I is incorrect but Statement II is true

12. Match List-I with List-II.

[NEET-2023-MANIPUR]

List-I

List-II

(A) Columnar epithelium

(I) Ducts of glands

(B) Ciliated epithelium

(II) Inner lining of stomach and intestine

(C) Squamous epithelium

(III) Inner lining of bronchioles

(D) Cuboidal epithelium

(IV) Endothelium

(1) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(2) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(3) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

(4) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

13. Given below are two statements:

[NEET-2023]

Statement I: Ligaments are dense irregular tissue.

Statement II: Cartilage is dense regular tissue.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I false is but Statement II is true

14. Which of the following is characteristic feature of cockroach regarding sexual dimorphism?

(1) Dark brown body colour and anal cerci

[NEET-2023]

(2) Presence of anal styles

(3) Presence of sclerites

(4) Presence of anal cerci

15. In cockroach, excretion is brought about by

[NEET-2023]

A. Phallic gland

B. Urecose gland

C. Nephrocytes

D. Fat body

E. Collateral glands

Choose the correct answer from the options given below :

(1) A and E only

(2) A, B and E only

(3) B, C and D only

(4) B and D only

16. Which of the following is not a connective tissue? [NEET-2022]

- | | |
|--------------------|---------------|
| (1) Adipose tissue | (2) Cartilage |
| (3) Neuroglia | (4) Blood |

17. Tegmina in cockroach, arises from: [NEET-2022]

- | | |
|------------------------------|----------------|
| (1) Mesothorax | (2) Metathorax |
| (3) Prothorax and Mesothorax | (4) Prothorax |

18. Which of the following types of epithelium is present in the bronchioles and Fallopian tubes?

- (1) Simple squamous epithelium [Re- NEET-2022]
(2) Simple columnar epithelium
(3) Ciliated epithelium
(4) Stratified squamous epithelium

19. Match List-I with List-II regarding the organs of Cockroach: [Re- NEET-2022]

- | List-I | List-II |
|------------------------|------------------------------------|
| (a) Crop | (i) grinding the food particles |
| (b) Proventriculus | (ii) secretion of digestive juice |
| (c) Hepatic caecae | (iii) removal of nitrogenous waste |
| (d) Malpighian tubules | (iv) storage of food |
- (1) (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)
(2) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)
(3) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)
(4) (a) - (i), (b) - (iv), (c) - (iii), (d) - (ii)

20. Which of the following characteristics is incorrect with respect to cockroach? [NEET-2021]

- (1) A ring of gastric caeca is present at the junction of midgut and hind gut.
(2) Hypopharynx lies within the cavity enclosed by the mouth parts.
(3) In females, 7th - 9th sterna together form a genital pouch.
(4) 10th abdominal segment in both sexes, bears a pair of anal cerci.

21. Goblet cells of alimentary canal are modified from : [NEET-2020]

- | | |
|-------------------------------|-------------------------------|
| (1) Compound epithelial cells | (2) Squamous epithelial cells |
| (3) Columnar epithelial cells | (4) Chondrocytes |

22. Cuboidal epithelium with brush border of microvilli is found in : [NEET-2020]

- (1) eustachian tube
(2) lining of intestine
(3) ducts of salivary glands
(4) proximal convoluted tubule of nephron

23. Select the incorrectly matched pair from following:

[NEET-2020]

- (1) Chondrocytes - Smooth muscle cells
- (2) Neurons - Nerve cells
- (3) Fibroblast - Areolar tissue
- (4) Osteocytes - Bone cells

24. If the head of cockroach is removed, it may live for few days because: -

[NEET-2020]

- (1) the supra-oesophageal ganglia of the cockroach are situated in ventral part of abdomen.
- (2) the cockroach does not have nervous system.
- (3) the head holds a small proportion of a nervous system while the rest is situated along the ventral part of its body.
- (4) the head holds a 1/3rd of a nervous system while the rest is situated along the dorsal part of its body.

25. In cockroach, identify the parts of the foregut in correct sequence: -

[NEET-2020(COVID-19)]

- (1) Mouth → Oesophagus → Pharynx → Crop → Gizzard
- (2) Mouth → Crop → Pharynx → Oesophagus → Gizzard
- (3) Mouth → Gizzard → Crop → Pharynx → Oesophagus
- (4) Mouth → Pharynx → Oesophagus → Crop → Gizzard

26. Match the following columns with reference to cockroach and select the correct option :

[NEET-2020(COVID-19)]

Column - I

Column - II

- | | |
|------------------------------------|----------------------|
| (a) Grinding of the food particles | (i) Hepatic caecal |
| (b) Secrete gastric juice | (ii) 10th segment |
| (c) 10 pairs | (iii) Proventriculus |
| (d) Anal cerci | (iv) Spiracles |
| | (v) Alary muscles |

- (1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (2) (a)-(iv), (b)-(iii), (c)-(v), (d)-(ii)
- (3) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
- (4) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

27. The ciliated epithelial cells are required to move particles or mucus in a specific direction. In humans, these cells are mainly present in :

[NEET-2019]

- (1) Bile duct and Bronchioles
- (2) Fallopian tubes and Pancreatic duct
- (3) Eustachian tube and Salivary duct
- (4) Bronchioles and Fallopian tubes

28. Select the correct sequence of organs in the alimentary canal of cockroach starting from mouth :-

- (1) Pharynx → Oesophagus → Crop → Gizzard → Ileum → Colon → Rectum [NEET-2019]
- (2) Pharynx → Oesophagus → Gizzard → Crop → Ileum → Colon → Rectum
- (3) Pharynx → Oesophagus → Gizzard → Ileum → Crop → Colon → Rectum
- (4) Pharynx → Oesophagus → Ileum → Crop → Gizzard → Colon → Rectum

29. Which of the following features is used to identify a male cockroach from a female cockroach?

- (1) Presence of a boat shaped sternum on the 9th abdominal segment [NEET-2018]
- (2) Presence of caudal styles or anal styles
- (3) Forewings with darker tegmina
- (4) Presence of anal cerci

30. Select the correct route for the passage of sperms in the male frogs. [NEET-2017]

- (1) Testis → Bidder's canal → Kidney → Vasa efferentia → Urinogenital duct → Cloaca
- (2) Testis → Vasa efferentia → Kidney → Seminal Vesicle → Urinogenital duct → Cloaca
- (3) Test → Vasa efferentia → Bidder's canal → Ureter → Cloaca
- (4) Testis → Vasa efferentia → Kidney → Bidder's canal → Urinogenital duct → Cloaca

31. Frog's heart when taken out of the body continues to beat for sometime. [NEET-2017]

Select the best option from the following statements.

I. Frog is a poikilotherm.

II. Frog does not have any coronary circulation.

III. Heart is "myogenic" in nature.

IV. Heart is autoexcitable.

- (1) Only III
- (2) Only IV
- (3) I and II
- (4) III and IV

32. Which type of tissue correctly matches with its location ? [NEET-I 2016]

- (1) Smooth muscle - Wall of intestine
- (2) Areolar tissue - Tendons
- (3) Transitional epithelium - Tip nose
- (4) Cuboidal epithelium - Lining of stomach

33. Connective tissues are derived from embryonic [NEET-I 2016]

- (1) Ectoderm
- (2) Endoderm
- (3) Endo mesoderm
- (4) Mesoderm

ANSWER KEY

1. (2)
2. (4)
3. (2)
4. (4)
5. (2)
6. (2)
7. (3)
8. (1)
9. (1)
10. (4)
11. (2)
12. (2)
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23. (1)
24. (4)
25. (4)
26. (1)
27. (4)
28. (1)
29. (2)
30. (2)
31. (4)
32. (1)
33. (4)